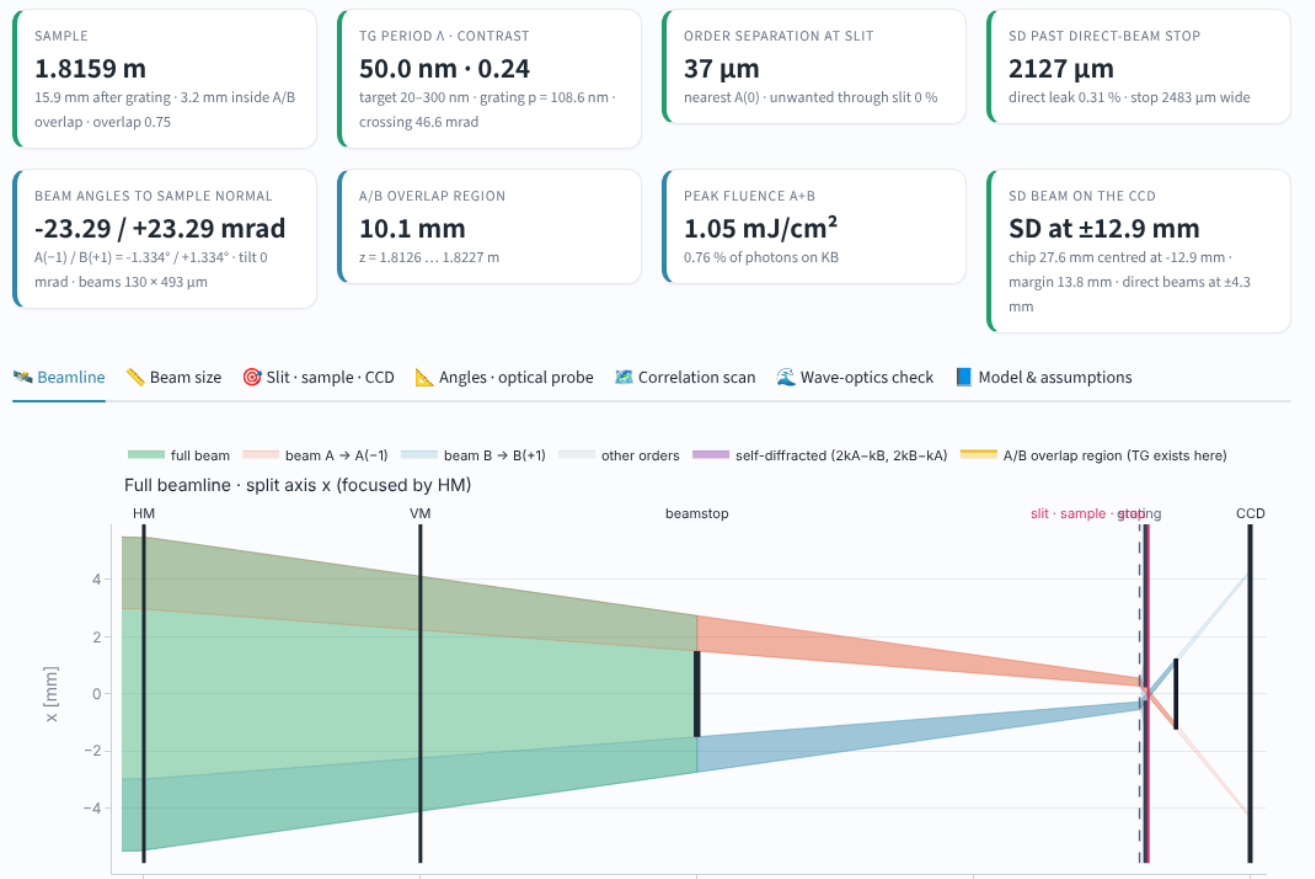
SD-XTG beam-path simulator

What the GUI does and how we use it · PAL beamtime, O K-edge (λ = 2.33 nm) · D. Fainozzi, October 2026

What it is for

We want a transient grating made by the FEL itself. The KB focuses the beam (H mirror f = 2 m, V mirror 0.5 m downstream, f = 1.5 m, ~40 µm focus on the Princeton CCD). A bar beamstop after the KB cuts the beam into two converging half-beams, **A** and **B**. A grating diffracts both, and the 1st orders **A(−1)** and **B(+1)** cross on the sample, where they write the TG. A slit just before the sample keeps only these two orders. Then either the self-diffracted X-rays ($2k_A - k_B$, $2k_B - k_A$) go to the CCD past a direct-beam stop (**SD mode**), or an optical beam probes the grating (**optical-probe mode**).

The simulator traces this beam path for the pink (4 mm) and mono (1 mm) beams, tells us where the sample has to be, which grating period gives the TG period we want, and whether every element (slit, beamstops, CCD, probe) still works. Every number in the window is recomputed when a parameter changes.



Main window, pink beam, Λ = 50 nm target. Status cards on top (green = fine, red = a condition fails), the beamline sketch below: full beam (green), A and B, all diffraction orders, slit, sample, overlap region (yellow) and direct-beam stop.

The constraint that drives everything: the TG period

We set a **target range** Λ_{min}–Λ_{max} in the sidebar (min = max for one exact period; goal: tens to hundreds of nm). Any setting whose TG period falls outside it is **discarded**: the card turns red, and the scans leave that pixel grey. In the default mode the grating period is not typed by hand: it is **solved** from the Λ setpoint with the exact grating equation sin β = sin α ± λ/p.

Two things come out of the geometry. First, $\Lambda = \lambda / (2 \sin(\theta/2))$, θ = angle between A(−1) and B(+1), and $\theta = 2\lambda/p + \theta_{KB}$: the grating period sets Λ almost alone, the grating position only moves the sample ($R_{sample} = R_{grating} \cdot (2\lambda/p)/\theta$). Second, the largest Λ we can get is the one of A and B without a grating, from the KB convergence only (~630 nm pink, ~2.5 µm mono).

Λ target	grating period	crossing angle θ	beams to sample normal	grating → sample (pink, z_g = 1.80 m)
20 nm	41 nm	116 mrad	±58 mrad (±3.3°)	6 mm
50 nm	108.6 nm	46.6 mrad	±23.3 mrad (±1.33°)	16 mm
300 nm	1141 nm	7.7 mrad	±3.9 mrad (±0.22°)	95 mm
497 nm (old optimum)	4579 nm	4.69 mrad	±2.35 mrad (±0.134°)	418 mm (z_g = 1.467 m)

The old 3.6–4.6 µm numbers were grating periods, not TG periods: with λ = 2.33 nm the TG period comes from the crossing angle, and ~500 nm needs only ±2.35 mrad. Small Λ means large angles, so the SD beams leave at ~1.5 θ and miss a CCD on the axis; the code checks the chip (27.6 mm) and can move it sideways onto the SD beam.

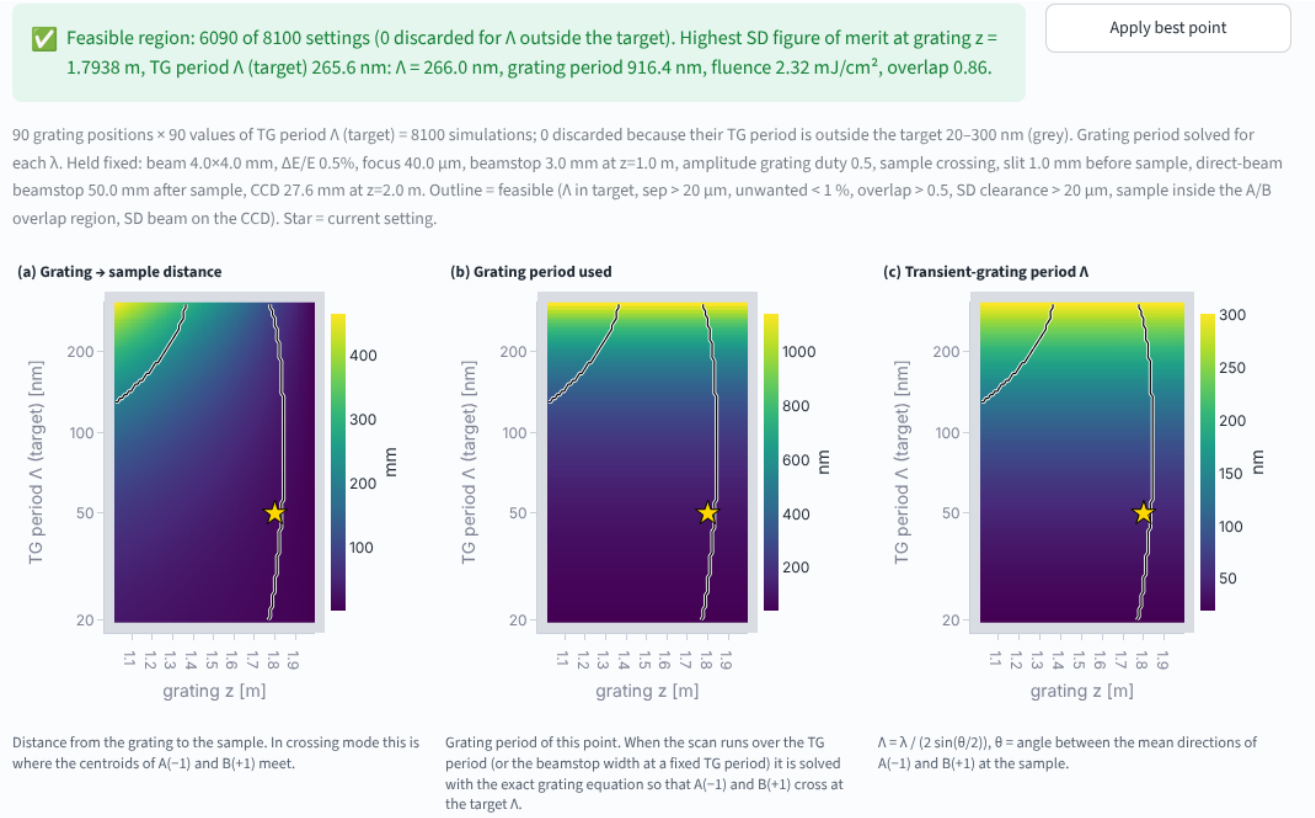
How we use it

1. **Start it.** `python sdxtg_app.py`, or open it in Spyder and press Run (F5). The browser opens on the GUI.
2. **Pick the beam** (pink / mono) and the **experiment** (SD on CCD / X-ray pump · optical probe).
3. **Set the TG target** range and the Λ setpoint. The sidebar prints the solved grating period.
4. **Move the elements**: beamstop z and width, grating z, type and duty cycle, slit distance, direct-beam stop, CCD. The cards tell us what fails.
5. **Run a correlation scan** (grating z × Λ, or × beamstop width for one exact Λ). The outlined region is feasible; click a pixel or Apply best point to load it.
6. **Check with waves**: the wave-optics tab propagates the field for the current setting and compares profiles, fringes and contrast with the ray model.
7. **Save** the parameters (JSON), the beam-size table and the scan (CSV).

Tab	What we see
Beamline	Full sketch HM → CCD and a zoom grating → CCD with every order, slit, sample, A/B overlap region, stops.
Beam size	FWHM vs z on both axes for the full beam, A and B; table of sizes at every element (CSV).
Slit · sample · CCD	Profiles at the slit (before the blades), at the sample, at the direct-beam stop, spots on the CCD.
Angles · optical probe	Angles of A(−1), B(+1) to the sample normal, crossing angle, Λ; probe phase matching (sinc^2 curve).
Correlation scan	Up to 250 × 250 full simulations, 12 maps, each with its definition printed under it.
Wave-optics check	Fresnel propagation: wave vs ray profiles, TG fringes, contrast, x–z intensity map.
Model & assumptions	Geometry, formulas and the placeholders to replace with real values.

Correlation scans

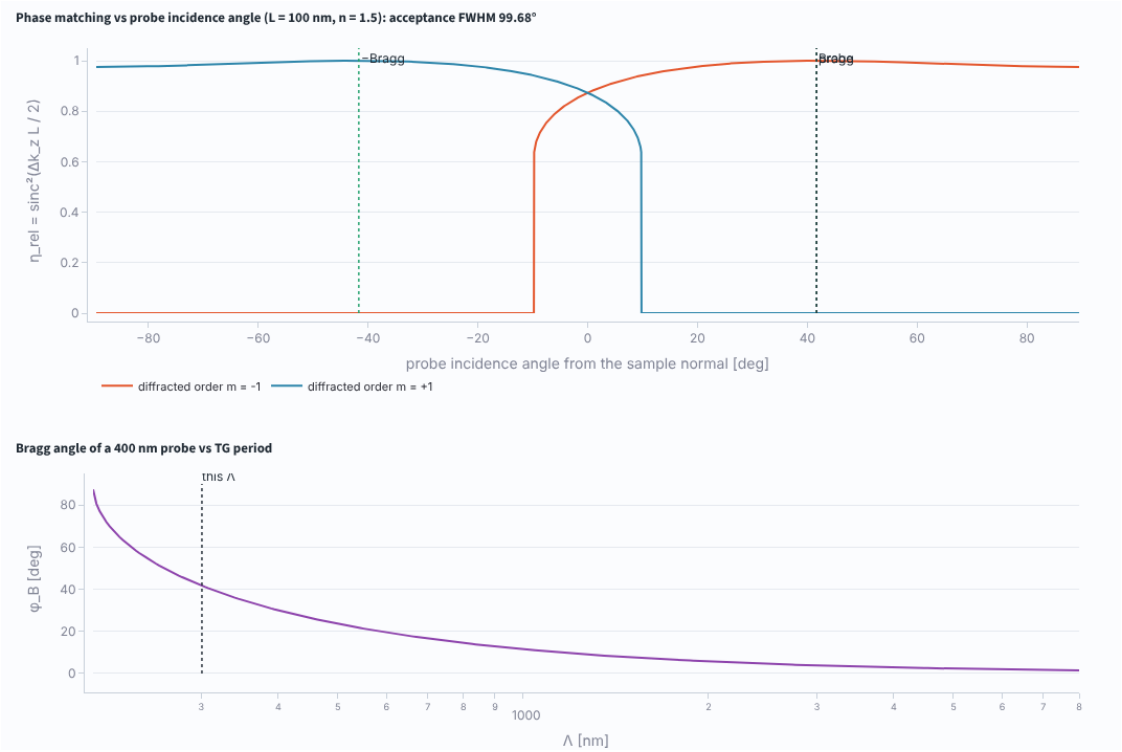
Each pixel is a full simulation; everything not on the axes is taken from the sidebar. The outline is the feasible region: Λ in target, sample inside the A/B overlap region, ≥ 20 µm between wanted and unwanted orders at the slit, < 1 % unwanted light, overlap > 0.5, and either SD clearance > 20 µm with the SD beam on the chip (SD mode) or $\Lambda > \lambda_{probe}/2$ (optical mode).



Correlation scan in the GUI, pink beam, grating z × Λ over 20–300 nm (90 × 90). Hover shows Λ, the solved grating period and the map value; a click applies the point.

Optical-probe mode

For an X-ray pump / optical probe experiment we set the probe wavelength, the sample index n, its thickness and the X-ray attenuation length (the TG is only as deep as the pump goes: L = min of the two). The probe keeps its in-plane wavevector plus ±2π/Λ exactly; the mismatch along the normal gives $\eta_{rel} = \mathrm{sinc}^2(\Delta k_z L/2)$. The tab plots the full curve vs incidence angle for both orders, the Bragg angle $\sin\varphi_B = \lambda_{probe}/(2\Lambda)$, the acceptance and the exit angle. A diffracted probe needs $\Lambda > \lambda_{probe}/2$ (200 nm for a 400 nm probe).



Optical-probe tab, Λ = 300 nm, 400 nm probe: $\varphi_B = 41.7°$. A 100 nm thick grating is in the Raman–Nath regime (very broad curve); a few µm thick one becomes angle-selective (5.5° FWHM at 5 µm).

How the physics is done

- **Ray model** (all interactive numbers and scans): rays with a mirror position, a focal-plane position (the 40 µm spot) and a wavelength; exact grating equation at the grating; efficiencies from the binary-grating Fourier coefficients (amplitude or phase grating). Mirror-symmetric sampling, so A and B are exact mirror images.
- **Wave model** (check tab): scalar Fresnel propagation along the split axis with the Fresnel scaling theorem for the converging beam; each grating order carried as envelope × exact carrier and summed coherently at the sample on a Λ/10 grid; partial coherence as a Gaussian Schell model; bandwidth as an incoherent sum. It works down to Λ ≈ 15 nm.
- **They agree**: TG contrast 0.24 vs 0.24 (pink, 50 nm), 0.24 vs 0.235 (20 nm), 0.24 vs 0.23 (250 nm); Λ to 0.1 %.
- **Contrast** is limited by the mutual coherence of A and B (they leave the KB ~7 mm apart for pink), not by the bandwidth: the TG is achromatic in this geometry.

What we still have to put in

- Measured PAL-XFEL transverse coherence (`coh_ratio`, now 2.7), bandwidth and pulse energy (placeholders).
- The focal-length convention (f_V from the VM or from the HM) and the CCD chip size and position.
- Probe wavelength, sample index, thickness and the X-ray attenuation length at 532 eV.
- Real grating: bar transmission/phase at 532 eV; a π phase grating gives ~4× the 1st-order efficiency of an opaque one.

Code: github.com/D3nFaina/xtg_and_sdxtg-simulator · MIT licence · files `sdxtg_sim.py` (physics, scans, command line), `sdxtg_wave.py` (wave optics), `sdxtg_app.py` (GUI).